2.49 Probability: Conditional Probability

Worked Solutions

1)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\frac{1}{10}}{\frac{1}{2}} = \frac{1}{5}$

2)

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

$\Rightarrow \frac{1}{2} = \frac{1}{4} + \frac{2}{5} - \mathrm{P}(A \cap B) \Rightarrow \mathrm{P}(A \cap B) = \frac{3}{20}$

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\frac{3}{20}}{\frac{1}{4}} = \frac{3}{5}$

3)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \Rightarrow 0.5 = \frac{0.3}{0.3+y} \Rightarrow y = 0.3$

$1 = 0.2 + x + 0.3 + 0.3 \Rightarrow x = 0.2$

4 a)

$0.4 = 0.22 + 0.15 + z \Rightarrow z = 0.03$

4 b)

$\mathrm{P}(B \cap C) = \mathrm{P}(B) \times \mathrm{P}(C)$

$\Rightarrow 0.22 = (0.25 + y + 0.22) \times 0.4$

$\Rightarrow y + 0.47 = 0.55 \Rightarrow y = 0.08$

4 c)

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} \Rightarrow 0.625 = \frac{0.25}{x+0.25} \Rightarrow x = 0.15$

4 d)

$1 = w + 0.15 + 0.25 + 0.08 + 0.22 + 0.15 + 0.03$

$\Rightarrow 1 = w + 0.88 \Rightarrow w = 0.12$

5)

$\mathrm{P}(\text{History}) = \frac{1}{4}$, $\mathrm{P}(\text{Geography}) = \frac{2}{5}$

$\mathrm{P}(\text{History} \mid \text{Geography}) = \frac{\mathrm{P}(\text{History} \cap \text{Geography})}{\mathrm{P}(\text{Geography})}$

$\Rightarrow \frac{1}{3} = \frac{\mathrm{P}(\text{History} \cap \text{Geography})}{\frac{2}{5}}$

$\Rightarrow \mathrm{P}(\text{History} \cap \text{Geography}) = \frac{2}{15}$

$\mathrm{P}(\text{History} \cup \text{Geography})$

$= \mathrm{P}(\text{History}) + \mathrm{P}(\text{Geography}) - \mathrm{P}(\text{History} \cap \text{Geography})$

$= \frac{1}{4} + \frac{2}{5} - \frac{2}{15} = \frac{31}{60}$

$\therefore \mathrm{P}((\text{History} \cup \text{Geography})') = 1 - \frac{31}{60} = \frac{29}{60}$

6)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \Rightarrow 0.3 = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \Rightarrow \mathrm{P}(B) = \frac{\mathrm{P}(A \cap B)}{0.3}$

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} \Rightarrow 0.4 = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} \Rightarrow \mathrm{P}(A) = \frac{\mathrm{P}(A \cap B)}{0.4}$

$\mathrm{P}(A \cup B) = 1 - \mathrm{P}(A' \cap B') = 1 - 0.1 = 0.9$

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

$\Rightarrow 0.9 = \frac{\mathrm{P}(A \cap B)}{0.4} + \frac{\mathrm{P}(A \cap B)}{0.3} - \mathrm{P}(A \cap B)$

$\Rightarrow 0.9 = \frac{29}{6}\mathrm{P}(A \cap B) \Rightarrow \mathrm{P}(A \cap B) = \frac{27}{145}$

7 a)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\frac{2}{7} \times \frac{4}{5}}{\frac{2}{7} \times \frac{4}{5} + \frac{5}{7} \times \frac{2}{3}} = \frac{12}{37}$

7 b)

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\frac{2}{7} \times \frac{4}{5}}{\frac{2}{7}} = \frac{4}{5}$

7 c)

$\mathrm{P}(A \mid B') = \frac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = \frac{\frac{2}{7} \times \frac{1}{5}}{\frac{2}{7} \times \frac{1}{5} + \frac{5}{7} \times \frac{1}{3}} = \frac{6}{31}$

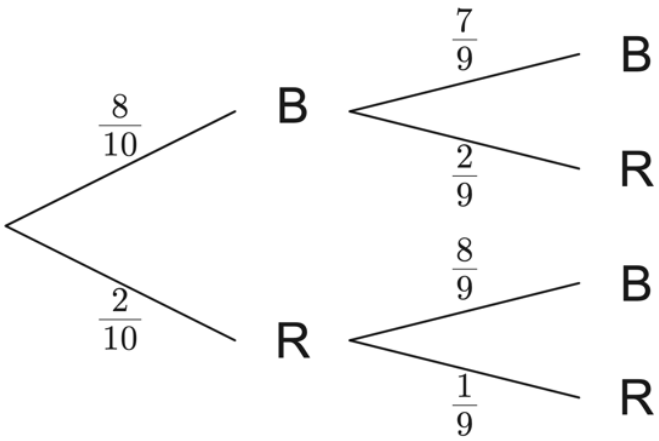
7 d)

$\mathrm{P}(B' \mid A) = \frac{\mathrm{P}(A \cap B')}{\mathrm{P}(A)} = \frac{\frac{2}{7} \times \frac{1}{5}}{\frac{2}{7}} = \frac{1}{5}$

7 e)

$\mathrm{P}(A' \mid B') = \frac{\mathrm{P}(A' \cap B')}{\mathrm{P}(B')} = \frac{\frac{5}{7} \times \frac{1}{5}}{\frac{2}{7} \times \frac{1}{5} + \frac{5}{7} \times \frac{1}{3}} = \frac{25}{31}$

8)



$\mathrm{P}(\text{1st Blue} \mid \text{2nd Red}) = \frac{\mathrm{P}(\text{1st Blue} \cap \text{2nd Red})}{\mathrm{P}(\text{2nd Red})}$

$= \frac{\frac{8}{10} \times \frac{2}{9}}{\frac{8}{10} \times \frac{2}{9} + \frac{2}{10} \times \frac{1}{9}} = \frac{8}{9}$

9 a)

$\mathrm{P}(\text{plain milk chocolate}) = \frac{16}{60} = \frac{4}{15}$

9 b)

$\mathrm{P}(\text{plain chocolate}) = \frac{28}{60} = \frac{7}{15}$

9 c)

$\mathrm{P}(\text{white chocolate} \mid \text{plain chocolate}) = \frac{6}{28} = \frac{3}{14}$

9 d)

$\mathrm{P}(\text{hazelnut chocolate} \mid \text{white chocolate}) = \frac{5}{15} = \frac{1}{3}$

9 e)

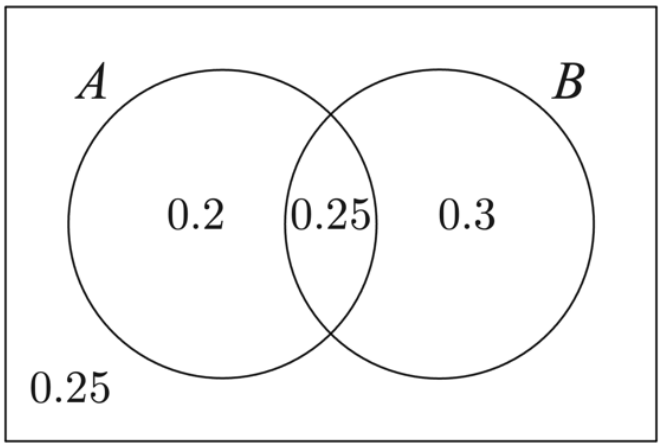
Any one of: Milk chocolate and Dark chocolate, Milk Chocolate and White Chocolate, Plain and Caramel, Plain and Hazelnut, Caramel and Hazelnut

10 a)

$\mathrm{P}(A \cap B) = 0.25$, $\mathrm{P}(A' \cap B) = 0.3$

$\mathrm{P}(B) = \mathrm{P}(A \cap B) + \mathrm{P}(A' \cap B) = 0.55$

10 b)



$\mathrm{P}(A) = \mathrm{P}(B') = 1 - 0.55 = 0.45$

$\therefore \mathrm{P}(A \cap B') = \mathrm{P}(A) - \mathrm{P}(A \cap B) = 0.2$

$\mathrm{P}(A' \cap B') = 1 - 0.2 - 0.25 - 0.3 = 0.25$

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{0.25}{0.55} = \frac{5}{11}$

10 c)

$\mathrm{P}(A) = 0.45$, $\mathrm{P}(A \mid B) = \frac{5}{11}$

$0.45 \neq \frac{5}{11}$

So A and B are not independent events

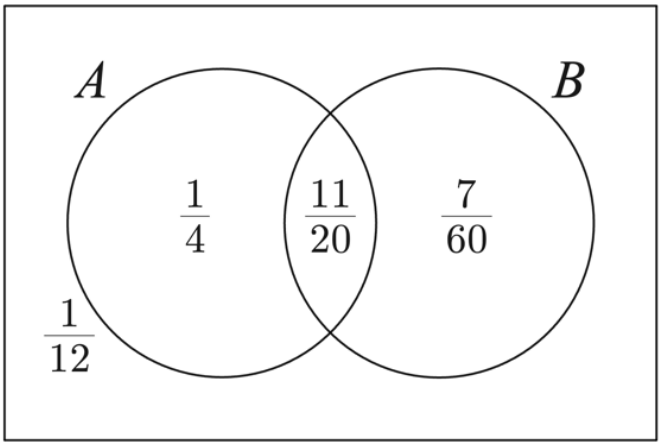
11 a)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\frac{11}{20}}{\frac{2}{3}} = \frac{33}{40}$

11 b)

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\frac{11}{20}}{\frac{4}{5}} = \frac{11}{16}$

11 c)



$\mathrm{P}(A \mid B') = \frac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}$

11 d)

$\mathrm{P}(B' \mid A') = \frac{\mathrm{P}(A' \cap B')}{\mathrm{P}(A')} = \frac{\frac{1}{12}}{\frac{1}{5}} = \frac{5}{12}$

11 e)

$\mathrm{P}([A \cup B]') = \frac{1}{12}$

12 a)

$\mathrm{P}(A \mid D) = \frac{\mathrm{P}(A \cap D)}{\mathrm{P}(D)} = \frac{\frac{1}{5} \times \frac{1}{2}}{\frac{1}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{7}{8} + \frac{2}{5} \times \frac{7}{10}}$

$= \frac{\frac{1}{10}}{\frac{1}{10} + \frac{7}{20} + \frac{7}{25}} = \frac{\frac{1}{10}}{\frac{73}{100}} = \frac{10}{73}$

12 b)

$\mathrm{P}(B \mid E) = \frac{\mathrm{P}(B \cap E)}{\mathrm{P}(E)} = \frac{\frac{2}{5} \times \frac{1}{8}}{\frac{1}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{1}{8} + \frac{2}{5} \times \frac{3}{10}}$

$= \frac{\frac{1}{20}}{\frac{1}{10} + \frac{1}{20} + \frac{3}{25}} = \frac{\frac{1}{20}}{\frac{27}{100}} = \frac{5}{27}$

12 c)

$\mathrm{P}(C \mid D) = \frac{\mathrm{P}(C \cap D)}{\mathrm{P}(D)} = \frac{\frac{2}{5} \times \frac{7}{10}}{\frac{1}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{7}{8} + \frac{2}{5} \times \frac{7}{10}}$

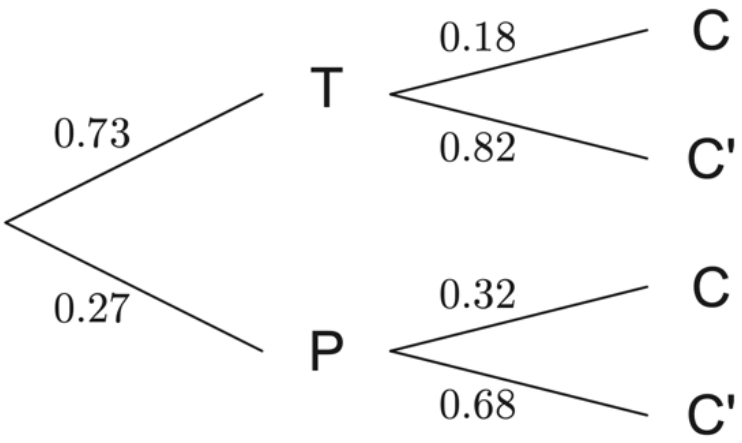
$= \frac{\frac{7}{25}}{\frac{1}{10} + \frac{7}{20} + \frac{7}{25}} = \frac{\frac{7}{25}}{\frac{73}{100}} = \frac{28}{73}$

12 d)

$\mathrm{P}(A' \mid E) = \frac{\mathrm{P}(A' \cap E)}{\mathrm{P}(E)} = \frac{\frac{2}{5} \times \frac{1}{8} + \frac{2}{5} \times \frac{3}{10}}{\frac{1}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{1}{8} + \frac{2}{5} \times \frac{3}{10}}$

$= \frac{\frac{1}{20} + \frac{3}{25}}{\frac{1}{10} + \frac{1}{20} + \frac{3}{25}} = \frac{\frac{17}{100}}{\frac{27}{100}} = \frac{17}{27}$

13 a)



$\mathrm{P}(T \mid C) = \frac{\mathrm{P}(T \cap C)}{\mathrm{P}(C)} = \frac{0.73 \times 0.18}{0.73 \times 0.18 + 0.27 \times 0.32} = \frac{0.1314}{0.2178} = \frac{73}{121}$

$= 0.6033$ (4 DP)

13 b)

$\mathrm{P}(P \mid C') = \frac{\mathrm{P}(P \cap C')}{\mathrm{P}(C')} = \frac{0.27 \times 0.68}{0.73 \times 0.82 + 0.27 \times 0.68} = \frac{0.1836}{0.7822} = \frac{918}{3911}$

$= 0.2347$ (4 DP)

14 a)

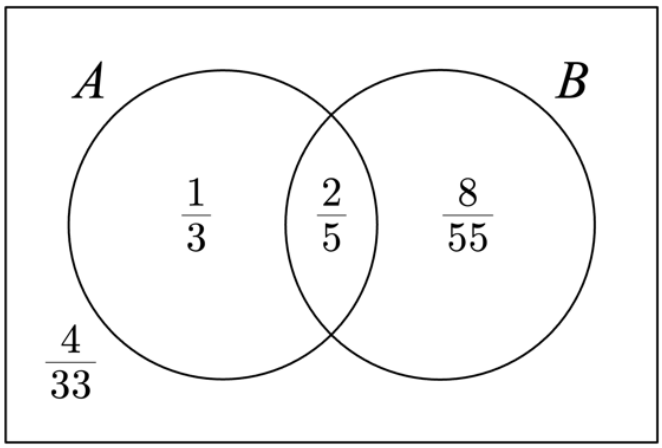
$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\frac{3}{5}}{\frac{2}{3}} = \frac{9}{10}$

14 b)

$\mathrm{P}(A) = \frac{7}{10} \neq \frac{9}{10} = \mathrm{P}(A \mid B)$

So A and B are not independent events.

15 a)



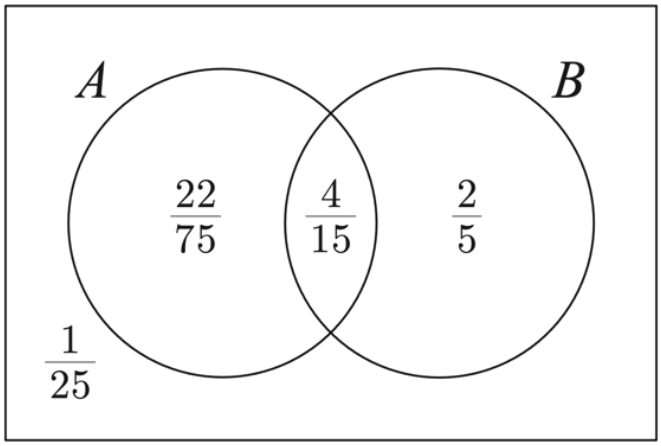
$\mathrm{P}(A' \mid B) = \frac{\mathrm{P}(A' \cap B)}{\mathrm{P}(B)} = \frac{\frac{8}{55}}{1-\frac{5}{11}} = \frac{4}{15}$

15 b)

$\mathrm{P}(A') = \frac{4}{15} = \mathrm{P}(A' \mid B)$

So A and B are independent events.

16)



$\mathrm{P}(A \mid [A \cap B]') = \frac{\mathrm{P}(A \cap [A \cap B]')}{\mathrm{P}([A \cap B]')} = \frac{\frac{22}{75}}{1-\frac{4}{15}} = \frac{2}{5}$

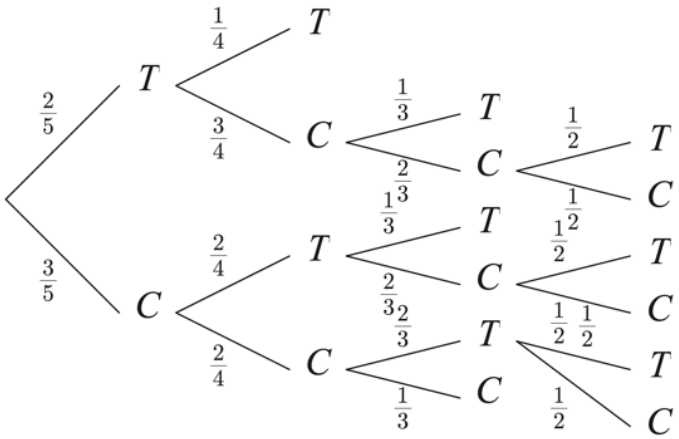
17)

	Fork	Knife	Spoon	Total
Silver	4	4	4	12
Silver-plated	6	6	6	18
Total	10	10	10	30

$\mathrm{P}(\text{Silver} \mid \text{Fork or Knife}) = \frac{\mathrm{P}(\text{Silver} \cap \text{Fork or Knife})}{\mathrm{P}(\text{Fork or Knife})}$

$= \frac{\frac{8}{30}}{\frac{20}{30}} = \frac{8}{20} = \frac{2}{5}$

18 a)



18 b)

$\mathrm{P}(TCC) + \mathrm{P}(CTT) + \mathrm{P}(CTC) + \mathrm{P}(CCT)$
$= \frac{2}{5} \times \frac{3}{4} \times \frac{2}{3} + \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}$
$= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = \frac{4}{5}$

18 c)

$\mathrm{P}(\text{Last is Tea} \mid \text{3 Mugs}) = \frac{\mathrm{P}(\text{Last is Tea} \cap \text{3 Mugs})}{\mathrm{P}(\text{3 Mugs})}$
$= \frac{\mathrm{P}(TCT)+\mathrm{P}(CTT)}{\mathrm{P}(TCT)+\mathrm{P}(CTT)+\mathrm{P}(CCC)}$
$= \frac{\frac{2}{5}\times\frac{3}{4}\times\frac{1}{3}+\frac{3}{5}\times\frac{2}{4}\times\frac{1}{3}}{\frac{2}{5}\times\frac{3}{4}\times\frac{1}{3}+\frac{3}{5}\times\frac{2}{4}\times\frac{1}{3}+\frac{3}{5}\times\frac{2}{4}\times\frac{1}{3}}$
$= \frac{\frac{1}{10}+\frac{1}{10}}{\frac{1}{10}+\frac{1}{10}+\frac{1}{10}} = \frac{\frac{2}{10}}{\frac{3}{10}} = \frac{2}{3}$

18 d)

$\mathrm{P}(\text{1st Tea} \mid \text{2nd Coffee}) = \frac{\mathrm{P}(\text{1st Tea} \cap \text{2nd Coffee})}{\mathrm{P}(\text{2nd Coffee})}$
$= \frac{\frac{2}{5}\times\frac{3}{4}}{\frac{2}{5}\times\frac{3}{4}+\frac{3}{5}\times\frac{2}{4}} = \frac{\frac{3}{10}}{\frac{3}{5}} = \frac{1}{2}$

19 a)

$\mathrm{P}(A \cap B) = \frac{30}{300} = \frac{1}{10}$

19 b)

$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\frac{30}{300}}{\frac{70}{300}} = \frac{30}{70} = \frac{3}{7}$

19 c)

$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\frac{30}{300}}{\frac{120}{300}} = \frac{30}{120} = \frac{1}{4}$

19 d)

$\mathrm{P}(C \mid B) = \frac{\mathrm{P}(B \cap C)}{\mathrm{P}(B)} = \frac{\frac{40}{300}}{\frac{70}{300}} = \frac{40}{70} = \frac{4}{7}$

19 e)

$\mathrm{P}(B \mid C) = \frac{\mathrm{P}(B \cap C)}{\mathrm{P}(C)} = \frac{\frac{40}{300}}{\frac{160}{300}} = \frac{40}{160} = \frac{1}{4}$

19 f)

$\mathrm{P}(A \mid [B \cup C]) = \frac{\mathrm{P}(A \cap [B \cup C])}{\mathrm{P}([B \cup C])} = \frac{\frac{85}{300}}{\frac{190}{300}} = \frac{85}{190} = \frac{17}{38}$

19 g)

$\mathrm{P}(B \mid [A \cup C]) = \frac{\mathrm{P}(B \cap [A \cup C])}{\mathrm{P}([A \cup C])} = \frac{\frac{60}{300}}{\frac{215}{300}} = \frac{60}{215} = \frac{12}{43}$

19 h)

$\mathrm{P}(B \mid [A \cap C]) = \frac{\mathrm{P}(A \cap B \cap C)}{\mathrm{P}([A \cap C])} = \frac{\frac{10}{300}}{\frac{65}{300}} = \frac{10}{65} = \frac{2}{13}$

19 i)

$\mathrm{P}([B \cup C] \mid A) = \frac{\mathrm{P}([B \cup C] \cap A)}{\mathrm{P}(A)} = \frac{\frac{85}{300}}{\frac{120}{300}} = \frac{85}{120} = \frac{17}{24}$

19 j)

$\mathrm{P}([B \cap C] \mid A) = \frac{\mathrm{P}(A \cap B \cap C)}{\mathrm{P}(A)} = \frac{\frac{10}{300}}{\frac{120}{300}} = \frac{10}{120} = \frac{1}{12}$

19 k)

$\mathrm{P}([A \cap B] \mid [A \cup B]) = \frac{\mathrm{P}([A \cap B] \cap [A \cup B])}{\mathrm{P}([A \cup B])} = \frac{\frac{30}{300}}{\frac{160}{300}} = \frac{30}{160} = \frac{3}{16}$

20 a)

	Year12	Year13	Total
Psychology	250	200	450
Physics	100	80	180
Chemistry	60	40	100
Total	410	320	730

58% of $450 = 261$ Psychology students study Maths

95% of $180 = 171$ Physics students study Maths

85% of $100 = 85$ Chemistry students study Maths

So there are a total of 517 students studying Maths

$\{A \cap C\} = 80$ as there are 80 Year13 Physics students.

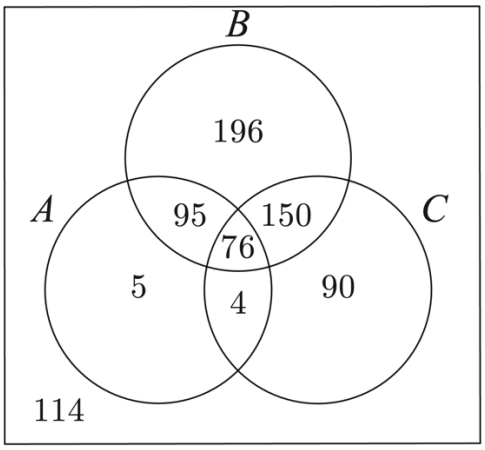
$\{A \cap B\} = 171$ students who study both Maths and Physics

$\therefore \{A \cap B \cap C\} = 76$ as $95 + 76 = 171$

$\Rightarrow \{B' \cap [A \cap C]\} = 4$ as $76 + 4 = 80$

$\{B \cap [A \cup C]'\} = 196$ as $517 - 95 - 76 - 150 = 196$

$\{[A \cup B \cup C]'\} = 730 - (196 + 95 + 76 + 150 + 5 + 4 + 90) = 114$



20 b)

$\mathrm{P}(A \cap B \cap C) = \frac{76}{730} = \frac{38}{365} = 0.1041$ (4 DP)

20 c)

$\mathrm{P}(A' \cap B \cap C) = \frac{150}{730} = \frac{15}{73} = 0.2055$ (4 DP)

20 d)

$\mathrm{P}([A \cup B] \mid [B \cup C]) = \frac{\mathrm{P}([A \cup B] \cap [B \cup C])}{\mathrm{P}(B \cup C)}$

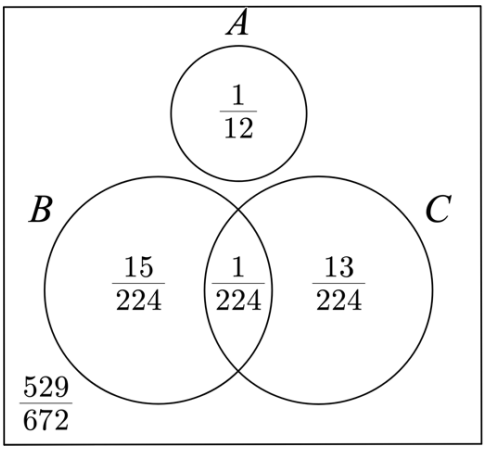
$= \frac{95+76}{196+95+76+150+4+90} = \frac{171}{611} = 0.2799$ (4 DP)

20 e)

$\mathrm{P}([A \cap B] \mid [B \cap C]) = \frac{\mathrm{P}([A \cap B] \cap [B \cap C])}{\mathrm{P}(B \cap C)} = \frac{76}{76+150} = \frac{38}{113}$

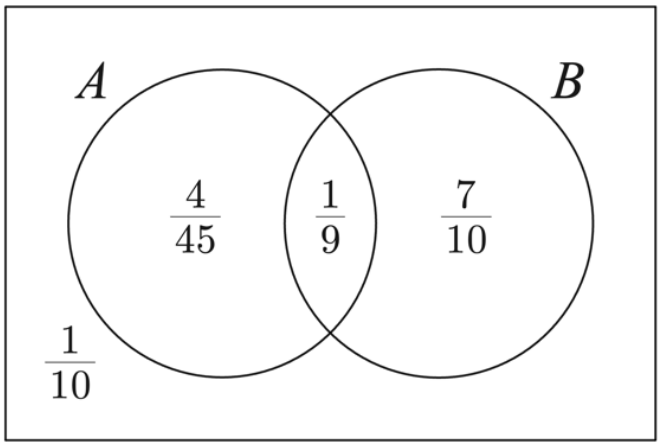
$= 0.3363$ (4 DP)

21)



$\mathrm{P}([A \cup B] \mid [B \cup C]) = \frac{\mathrm{P}([A \cup B] \cap [B \cup C])}{\mathrm{P}(B \cup C)} = \frac{\mathrm{P}(B)}{\mathrm{P}(B \cup C)} = \frac{\frac{1}{14}}{\frac{29}{224}} = \frac{16}{29}$

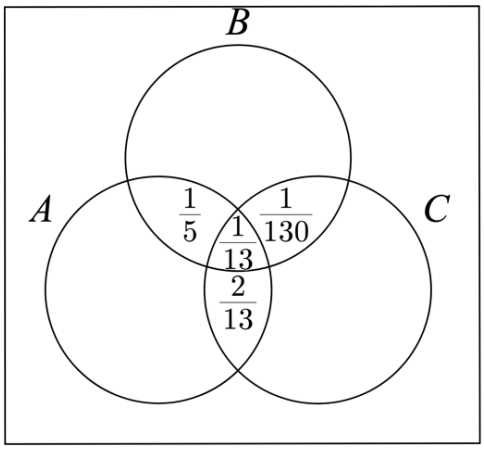
22)



$\mathrm{P}([A \cup B] \mid [A \cup B']) = \frac{\mathrm{P}([A \cup B] \cap [A \cup B'])}{\mathrm{P}([A \cup B'])} = \frac{\frac{4}{45}+\frac{1}{9}}{\frac{4}{45}+\frac{1}{9}+\frac{1}{10}} = \frac{\frac{1}{5}}{\frac{3}{10}} = \frac{2}{3}$

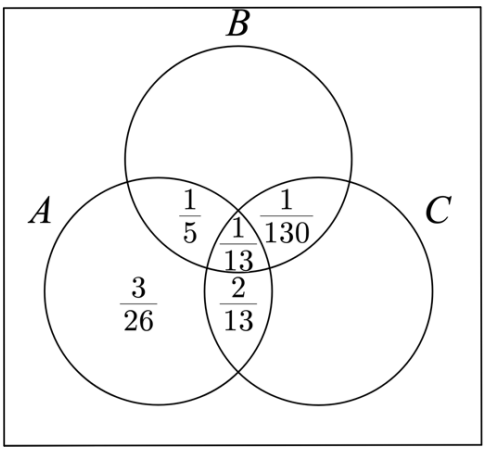
23)

$\mathrm{P}(A \cap B \cap C) = \frac{1}{13}$, $\mathrm{P}(A \cap B) = \frac{18}{65}$, $\mathrm{P}(A \cap C) = \frac{3}{13}$ and $\mathrm{P}(B \cap C) = \frac{11}{130}$ give:



$\mathrm{P}(B \mid A) = \frac{36}{71} \Rightarrow \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{36}{71} \Rightarrow \mathrm{P}(A \cap B) = \frac{36}{71}\mathrm{P}(A)$

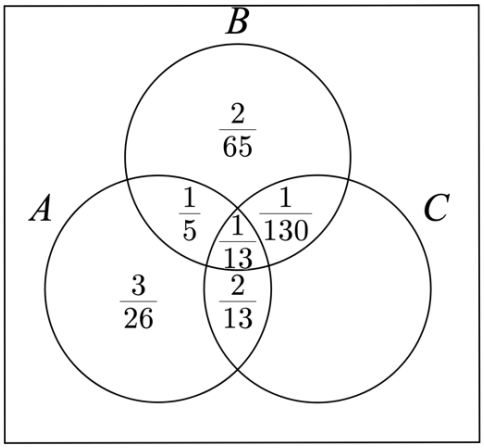
$\Rightarrow \frac{18}{65} = \frac{36}{71}\mathrm{P}(A) \Rightarrow \mathrm{P}(A) = \frac{71}{130}$



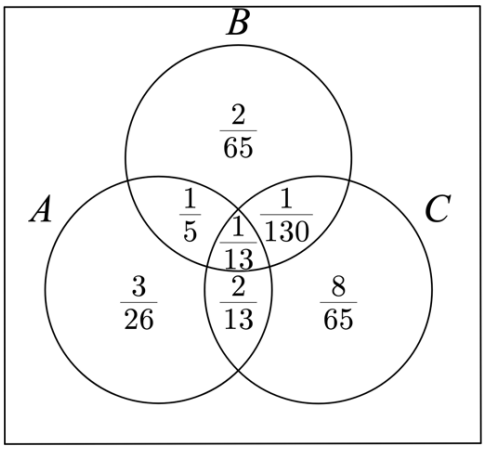
$\mathrm{P}(A \mid B) = \frac{36}{41} \Rightarrow \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{36}{41} \Rightarrow \mathrm{P}(A \cap B) = \frac{36}{41}\mathrm{P}(B)$

$\Rightarrow \frac{18}{65} = \frac{36}{41}\mathrm{P}(B) \Rightarrow \mathrm{P}(B) = \frac{41}{130}$

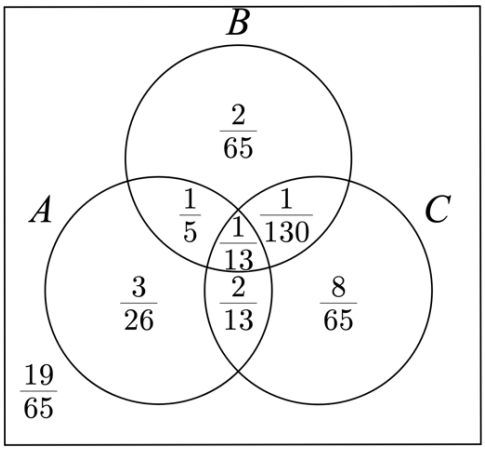
$\mathrm{P}(B \cap [A \cup C]') = \frac{41}{130} - \frac{1}{5} - \frac{1}{13} - \frac{1}{130} = \frac{2}{65}$



$\therefore \mathrm{P}(C \cap [A \cup B]') = 4 \times \frac{2}{65} = \frac{8}{65}$



$\mathrm{P}([A \cup B \cup C]') = 1 - \frac{2}{65} - \frac{1}{5} - \frac{1}{13} - \frac{1}{130} - \frac{3}{26} - \frac{2}{13} - \frac{8}{65} = \frac{19}{65}$



24)

$\mathrm{P}(A \mid B) = \frac{4}{11} \Rightarrow \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{4}{11} \Rightarrow \mathrm{P}(A \cap B) = \frac{4}{11}\mathrm{P}(B)$

$\mathrm{P}(B \mid A) = \frac{40}{49} \Rightarrow \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{40}{49} \Rightarrow \mathrm{P}(A \cap B) = \frac{40}{49}\mathrm{P}(A)$

$\therefore \frac{4}{11}\mathrm{P}(B) = \frac{40}{49}\mathrm{P}(A) \Rightarrow \mathrm{P}(B) = \frac{110}{49}\mathrm{P}(A)$

$\mathrm{P}([A \cup B]') = \frac{3}{20} \Rightarrow \mathrm{P}(A \cup B) = \frac{17}{20}$

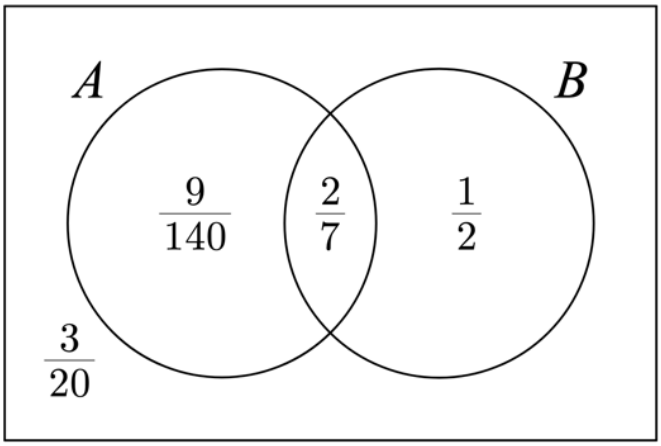
$\Rightarrow \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B) = \frac{17}{20}$

$\Rightarrow \mathrm{P}(A) + \frac{110}{49}\mathrm{P}(A) - \frac{40}{49}\mathrm{P}(A) = \frac{17}{20} \Rightarrow \frac{17}{7}\mathrm{P}(A) = \frac{17}{20}$

$\Rightarrow \mathrm{P}(A) = \frac{7}{20}$

$\therefore \mathrm{P}(B) = \frac{110}{49} \times \frac{7}{20} = \frac{11}{14}$

$\mathrm{P}(A \cap B) = \frac{4}{11} \times \frac{11}{14} = \frac{2}{7}$



25)

$\mathrm{P}(\text{hits bullseye at 200 cm}) = \frac{1}{6}$

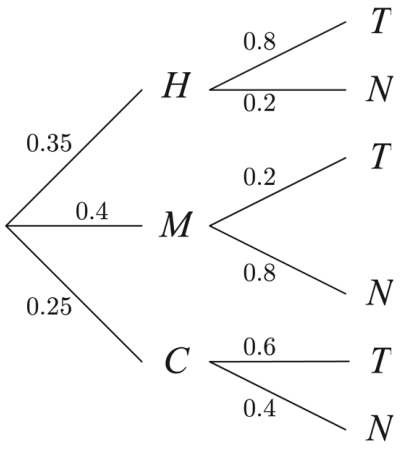
$\mathrm{P}(\text{hits bullseye at 210 cm}) = \frac{1}{6} \times 0.85 = \frac{17}{120}$

$\mathrm{P}(\text{hits bullseye at 220 cm}) = \frac{1}{6} \times 0.85^2 = \frac{289}{240}$

$\mathrm{P}(\text{hits bullseye at 230 cm}) = \frac{1}{6} \times 0.85^3 = \frac{4913}{48000}$

$= 0.1024$ (4 DP)

26)



$\mathrm{P}(H \mid T) = \frac{\mathrm{P}(H \cap T)}{\mathrm{P}(T)} = \frac{0.35 \times 0.8}{0.35 \times 0.8 + 0.4 \times 0.2 + 0.25 \times 0.6} = \frac{0.28}{0.51}$

$= \frac{28}{51} = 0.5490$ (4 DP)

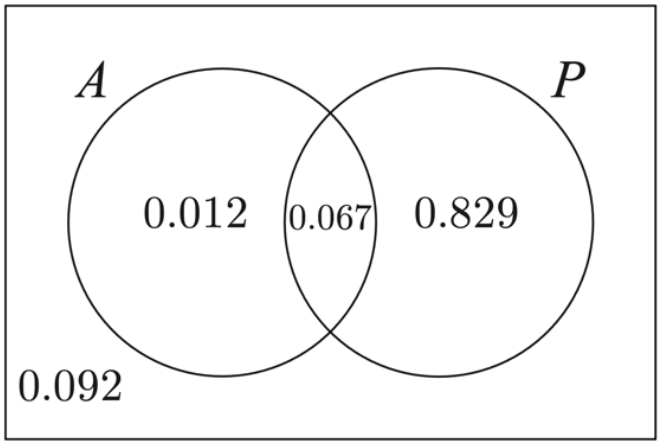
27)

$\mathrm{P}(\text{Have} \cup \text{Participated}) = 0.908$

$\mathrm{P}(\text{Have} \cup \text{Participated}) = \mathrm{P}(\text{Have}) + \mathrm{P}(\text{Participated}) - \mathrm{P}(\text{Have} \cap \text{Participated}) \Rightarrow$

$\mathrm{P}(\text{Have} \cap \text{Participated}) = 0.079 + 0.896 - 0.908$

$= 0.067$



$\mathrm{P}(\text{Have} \mid \text{Participated}) = \frac{\mathrm{P}(\text{Have} \cap \text{Participated})}{\mathrm{P}(\text{Participated})} = \frac{0.067}{0.896}$

$= \frac{67}{896} = 0.0748$ (4 DP)

28 a)

	Red	Green	Blue	Total
Gel	3	1	1	5
Felt-tip	4	2	2	8
Total	7	3	3	13

$\frac{7}{13} \times \frac{6}{12} \times \frac{5}{11} \times \frac{4}{10} = \frac{7}{143}$

28 b)

$\mathrm{P}(GGGF) + \mathrm{P}(GGFG) + \mathrm{P}(GFGG) + \mathrm{P}(FGGG)$

$= 4 \times \frac{5}{13} \times \frac{4}{12} \times \frac{3}{11} \times \frac{8}{10} = \frac{16}{143}$

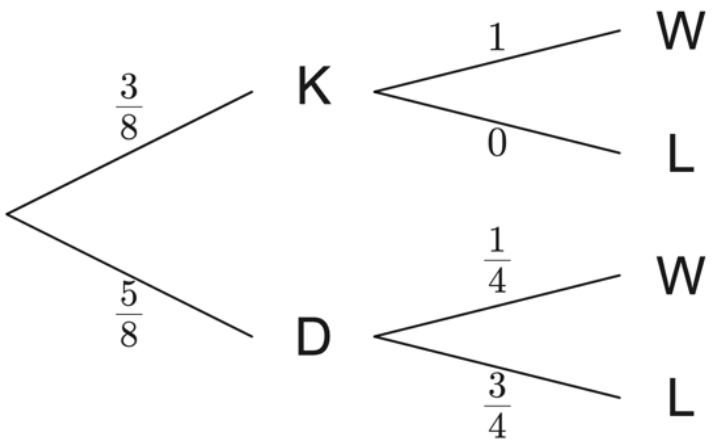
28 c)

$\mathrm{P}(\text{2 Felt-tip} \mid \text{3 Blue}) = \frac{\mathrm{P}(\text{2 Felt-tip} \cap \text{3 Blue})}{\mathrm{P}(\text{3 Blue})}$

$= \frac{\frac{2}{13} \times \frac{1}{12} \times \frac{1}{11} \times \frac{4}{10}}{\frac{3}{13} \times \frac{2}{12} \times \frac{1}{11} \times \frac{10}{10}} = \frac{2}{15}$

29)

The first question has the following probability tree, where K is the event that she knows the answer, D is the event that she does not know the answer, W is the event that she wins a question, and L is the event that she loses a question.



$\mathrm{P}(K \text{ Q1}) = \frac{3}{8}$

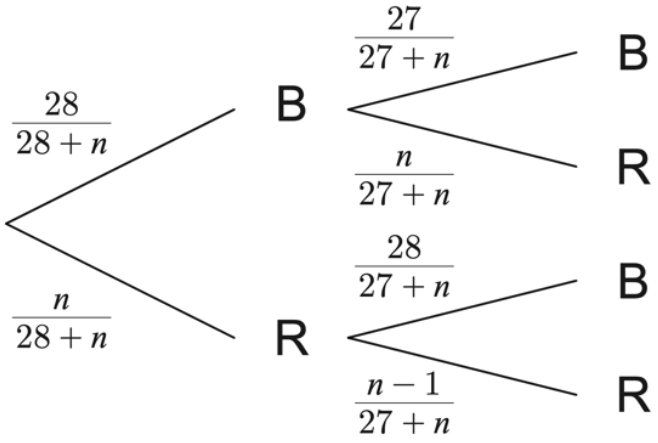
$\mathrm{P}(K \text{ Q2}) = \frac{3}{8} \times 0.9 = \frac{27}{80}$

$\mathrm{P}(K \text{ Q3}) = \frac{3}{8} \times 0.9^2 = \frac{243}{800}$

$\mathrm{P}(D \text{ Q3} \mid W \text{ Q3}) = \frac{\mathrm{P}(D \text{ Q3} \cap W \text{ Q3})}{\mathrm{P}(W \text{ Q3})} = \frac{\frac{17}{32} \times \frac{161}{320} \times (\frac{557}{800} \times \frac{1}{4})}{\frac{17}{32} \times \frac{161}{320} \times \frac{1529}{3200}}$

$= \frac{557}{1529} = 0.3643$ (4 DP)

30)



$\mathrm{P}(\text{Same Colour}) = \mathrm{P}(BB) + \mathrm{P}(RR) = 0.5$

$\Rightarrow \frac{28}{28+n} \times \frac{27}{27+n} + \frac{n}{28+n} \times \frac{n-1}{27+n} = \frac{1}{2}$

$\Rightarrow 756 + n(n-1) = \frac{1}{2}(28+n)(27+n)$

$\Rightarrow 1512 + 2n(n-1) = (28+n)(27+n)$

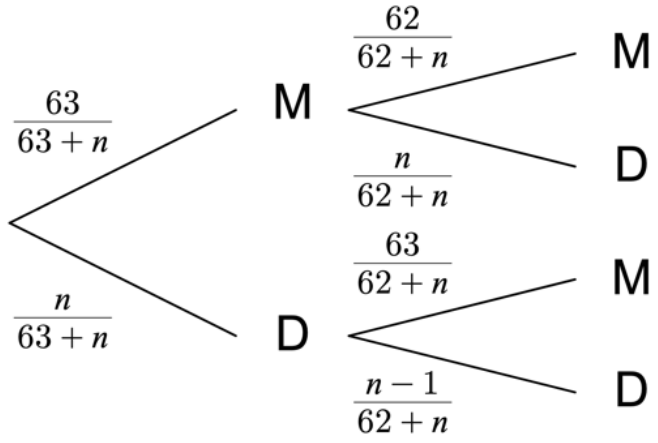
$\Rightarrow 1512 + 2n^2 - 2n = 756 + 55n + n^2$

$\Rightarrow n^2 - 57n + 756 = 0 \Rightarrow n = 36 \text{ or } n = 21$

As $n < 28$, we must have $n = 21$

$\mathrm{P}(1B \mid 2R) = \frac{\mathrm{P}(1B \cap 2R)}{\mathrm{P}(2R)} = \frac{\frac{28}{49} \times \frac{21}{48}}{\frac{28}{49} \times \frac{21}{48} + \frac{21}{49} \times \frac{20}{48}} = \frac{7}{12}$

31)



$\mathrm{P}(\text{Different Type}) = \mathrm{P}(MD) + \mathrm{P}(DM) = 0.35$

$\Rightarrow \frac{63}{63+n} \times \frac{n}{62+n} + \frac{n}{63+n} \times \frac{63}{62+n} = \frac{7}{20}$

$\Rightarrow 126n = \frac{7}{20}(63+n)(62+n)$

$\Rightarrow 2520n = 7(63+n)(62+n)$

$\Rightarrow 2520n = 7(3906 + 125n + n^2)$

$\Rightarrow 2520n = 27342 + 875n + 7n^2$

$\Rightarrow 7n^2 - 1645n + 27342 = 0$

$\Rightarrow n^2 - 235n + 3906 = 0 \Rightarrow n = 18$ or $n = 217$

As $n > 63$, we must have $n = 217$

$\mathrm{P}(1M \mid 2M) = \frac{\mathrm{P}(1M \cap 2M)}{\mathrm{P}(2M)} = \frac{\frac{63}{280} \times \frac{62}{279}}{\frac{63}{280} \times \frac{62}{279} + \frac{217}{280} \times \frac{63}{279}} = \frac{2}{9}$